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GE Medical Systems

Direction 2240919-100

Revision 5

SmartScore

Applications Tips and Precautions

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

This Service Manual is available in English only.

If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.

Do not attempt to service the equipment unless this service manual has been consulted and is understood.

Failure to heed this Warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

AVERTISSEMENT

Ce Manuel de maintenance n'est disponible qu'en anglais.

Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.

Ne pas tenter d'intervention sur les équipements tant que le manuel Service n'a pas été consulté et compris.

Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

Dieses Kundendienst-Handbuch existiert nur in englischer Sprache.

Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.

Versuchen Sie nicht, das Gerät zu reparieren, bevor dieses Kundendienst-Handbuch nicht zu Rate gezogen und verstanden wurde.

Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch elektrische Schläge, mechanische oder sonstige Gefahren kommen.

VISO

Este Manual de Servicio sólo existe en inglés.

Si algún proveedor de servicios ajeno a GEMS solicita un idioma que no sea el inglés, es responsabilidad del cliente ofrecer un servicio de traducción.

No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.

La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

LANGUAGE

ATENÇÃO

Este Manual de Assistência Técnica só se encontra disponível em Inglês.

Se qualquer outro serviço de assistência técnica, que não a GEMS, solicitar estes manuais noutra idioma, é da responsabilidade do cliente fornecer os serviços de tradução.

Não tente reparar o equipamento sem ter consultado e compreendido este Manual de Assistência Técnica.

O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

AVVERTENZA

Il presente manuale di manutenzione è disponibile soltanto in inglese.

Se un addetto alla manutenzione esterno alla GEMS richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.

Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.

Non tenere conto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

このサービスマニュアルには英語版しかありません。

GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage. Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins. 14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment. Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

LEGAL NOTES

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REVISION HISTORY

Revision	Date	Reason for change
0	03/24/99	New Document
1.0	09/21/99	Updated for SmartScore 2.02./ Helios 1.1ME3/M3
2.0	01/31/00	Updated for SmartScore 2.0.7/Helios 1.1 M#
3.0	10/05/00	Updated for SmartScore 3.2.2 Patient Report
4.0	08/06/01	Updated for SmartScore 3.3.3
5.0	08/25/01	Updated for SmartScore 3.3.4

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NEW FEATURES VERSION 3.3

Loading Messages

A new message will appear while the SmartScore software is loading. This message notifies the user of the software load process. The message will complete three steps before it disappears. They are: initializing componeents, initializing links and loading images. The study screen will appear and the last message, loading images, will still be running. While the images are loading, the user can type in information on the study screen.



Multi-Vendor Images

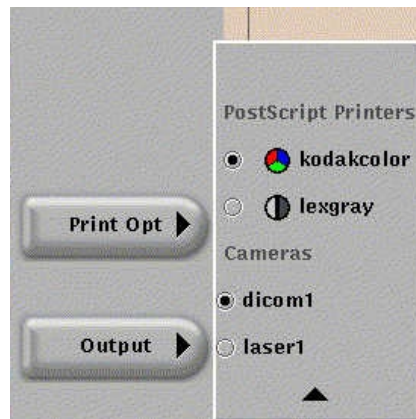
The SmartScore software now has the ability to score different vendors images. If the user has EBCT, Siemens, Toshiba, ect. DICOM images they can load them onto the AW system and then socre the images using the SmartScore software.

Caution: GE has not validated the scores by normal GE process

Print Opt

The Print Opt area allows the user to have multiple printers or cameras for printing films or patient reports with SmartScore. When the software is loaded, the system will detect the printers/cameras and a name will appear under the appropriate category. The name with the black circle is the printer or camera that will be used for the output. If the system has multiple printers/cameras, the user can select the printer/camera of choice by clicking on the circle next to the printer/camera. The last printer/camera that is selected before SmartScore is closed will be the printer/camera that is defaulted once SmartScore reopens.

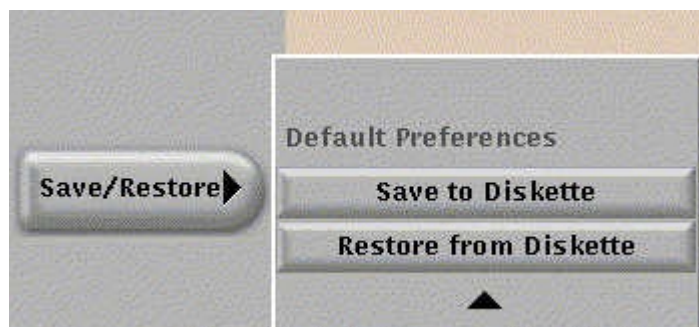
Note: If the printer/camera goes off line for some reason, it will be automatically removed from the Print Opt section. Please check the connection of the printer.



Save/Restore

The save/restore button allows the user to save or restore the information in the preference screens to diskette.

Once all of the information is loaded into the preference screens, place a floppy disk into the disk drive, click on the Save/Restore button and select Save to Diskette. The system will then save all of the preference information. Once the information has been stored, a message will appear telling the user the information has been saved. To restore the preference information, place the floppy disk into the disk drive and select Restore from Diskette. This will restore all of the preferences that were saved to the disk.



Warning Messages

A series of new warning message have been added to the SmartScore 3.3 software to help the user get accurate scores. The following messages may appear:

- **Some invalid images in this series have been discarded.**
This message will appear if there are corrupted images for some reason. The system will discard the bad images and allow the user to score the rest of the images.
- **The selected images cannot be used by the application due to inaccurate reconstruction type. The reconstruction type must be standard.**
This message will appear and disallow the user to score an image. The calcium scoring exam must be scanned with a standard reconstruction type in order to score the images. If another reconstruction mode is used, reconstruct the images again in the standard reconstruction mode.
- **Inaccurate score may result since DFOV is not 25cm**
This message will appear if a different DFOV was selected other than 25. A 25 DFOV is recommended to produce accurate scores. If another DFOV is used score values will change due to the pixel size being different than with a FOV of 25.
- **A Segmented reconstruction mode was not selected and this may cause low score values due to patient motion.**
This message will appear if a different reconstruction mode was used other than segment and may cause lower scores values due to motion. Reconstruct the images using a segmented reconstruction.
- **Inaccurate slice thickness has been selected which may result in inaccurate score. Recommended slice thickness is 3.0 mm or below. Please see operators manual for recommended slice thickness based on your scanner.**
This message will appear if a different slice thickness was used and lower scores may result due to partial volume affect. The user can score the images, but remember to use a 3mm slice thickness or lower. See operator manual for correct slice thickness of your scanner.
- **The selected images cannot be used by this application.**
This message will appear if inaccurate images were selected, eg. scout images. The software will not load the images.
- **Images are not from a GE scanner, Images are scorable by SmartScore, but score values have not been validated for this scanner by normal GE process.**

This message will appear when ever non-GE images are loaded into SmartScore.

WORKAROUNDS AND LIMITATIONS FOR LOAD PROCEDURE

1. If an incorrect license key is used, SmartScore will load correctly and will launch with only an error that states, "An Invalid License". The workaround is to reinstall the software.
2. Sometimes the software will not load successfully. If you look closely at the installation messages, there will be a message that a tar error has occurred. This seems to be a generic problem with Solaris when the CD-ROM does not mount successfully. The way to recover from this problem is to shutdown the AW and reboot. Immediately after the reboot, load the software.

WORKAROUNDS AND LIMITATIONS FOR SOFTWARE

1. If your printer goes off-line for some reason, the printer will be removed from the print option area and the user will not be able to print. Check the connection of the printer, or call service if the problem continues. If there are two printers the software will automatically default to the second printer.
2. The patient age field has now changed. The field will only allow the user to type in 3 characters of numbers only. The age field allows entry from 0 to 120. If an invalid age is entered into the age field and the user tries to screen save or print the patient report an invalid age message will appear. The user then needs to go back to the study screen and correct the age for these features to work.
3. Launching SmartScore without an EKG for CT/i image sets, causes the image set to be shifted to the right of the EKG display window. If this problem is seen, scroll the window to the right until the image diamonds become visible.
4. The Helical scan acquisition must not be longer than 60 seconds. This is due to a limitation in the maximum length of the EKG acquisition. For scans requiring more than 60 seconds increase pitch and reduce scan time to 60 seconds or less.
5. The SmartScore software needs the following dicom header information in order to start the software: Patient ID, Patient Name, Hospital name. If one of these fields is blank, the software will not open.
6. There is a problem when archiving screen saved SmartScore images created on the AW workstation using the CT Operator Console. When SmartScore screen saved images are created on the AW workstation from an image set created on a CT/i or X/i CT System, and then sent back to the CT scanner console the screen saved images can NOT be archived onto a MOD. If medical records are needed save the screen saved images to film. Another workaround is to display the Screen Saved images on the CT display and re-screen save the images.
7. Be patient with the software functionality; give the system time. When a command/job request (push button) is made, please wait for the command/job in progress to complete before selecting the next command/job.
8. Before running SmartScore ensure that 25% of the image disk is free (75% full) if not remove some of the exams before launching SmartScore.
9. If the operator can not avoid running SmartScore with the system above 90% full the following precautions should be observed:
 - Wait 30 seconds between output operations.
 - Output operations are defined as:
 - Save New Series, Save Screens, Send to Film, Send to Paper, Save Text
10. Avoid using the keyboard accelerator key, i.e. "Page Up/Page Down". Using these keys can corrupt the normal function of the menu buttons. If page up or page down are used, the arrows on the SmartScore software may disappear. To bring the arrows back, switch between the image and region screens.
11. International help screens show some figures that are annotated in English. The workaround is to reference the printed operator manual as the controlling document. The problems have been resolved in the printed version.
12. Please use the sort selection, "Sort by Number" when launching SmartScore. If other sort sections are used, the images will be displayed in reverse order in the SmartScore viewer screens.

13. The new “state save “ feature will only work in the full data set not the limited amount of saved axial images.

WORKAROUNDS AND LIMITATIONS FOR THE EKG MONITOR

1. The size of the R-Peak must be the same as or greater than the vertical scale line displayed the left of the EKG trace. The signal size can be increased by increasing the gain of the amplifier. Do not select the auto gain, select a numbered gain, i.e. 20mm/mv. (The default is 10mm/mV).
2. Do not eject the floppy disk prematurely.
3. For EKG acquisition always use formatted high density (1.44Mbyte) floppy disks. Do not format diskettes in UNIX machine. Use of diskettes preformatted for PC is recommended.
4. To avoid the possibility of the EKG monitor having a problem writing data to the floppy disk, always use a new formatted disk for each patient.
5. It is highly recommended that the power to the EKG monitor be turned OFF and then back ON between patients. This simple precautionary step could prevent EKG mis-timing problems at the beginning of the saved EKG waveform.
6. The EKG trace must have positive R-Peaks. For a negative R-Peak trace, switch the left and right EKG leads, i.e., swap "LA" with "RA" and "LL" with "RL".

PRECAUTIONS

1. Please pay close attention to verify a strong EKG trace signal on the EKG monitor before scanning is initiated. A trace must be visible before scanning be initiated otherwise NO trace will be available for the SmartScore analysis. Should the Left Arm (LA or Black lead) have a low input signal or if it falls off, the monitor will still continue to beep in a normal fashion and the Heart Rate will still be displayed BUT the EKG trace will not be displayed. Remember the EKG trace is needed for the monitor to save EKG data which is used for the analysis.
2. It is very important to properly place the EKG electrodes. Please refer to the SmartScore Operator's Manual.
3. With regards to breathing, using slow instructions have the patient take in a deep breath, blow it out, take in another deep breath and hold it
4. The scanning direction should always be from Superior to Inferior.
5. When saving the patient report to a floppy disk, do not remove the disk until the save is completed. A confirmation message will appear on the Windows workstation when the report is saved. If the disk is removed before the report is saved, the Advanced Windows workstation may crash.

PROPER SCORING

1. Start scoring at the superior most image demonstrating the Coronary vessels. Be careful of the LAD this artery will sometimes track superior from the Aortic root.
2. How large should the Region of Interest box be drawn when scoring calcification? The goal is to try to keep a consistent ratio within the Scoring Box of highlighted vs. non-highlighted pixels that will paint. A good rule to follow is to draw a box 50% larger than the area you are trying to score.
3. If artifact is associated with calcium (touching the calcium) then it is recommended that the artifact be included in the Scoring Box. If the artifact is not associated (does not touch the calcium) then the artifact should not be included in the Scoring Box. Following these simple rules will lead to more consistent scoring and increase repeatable scores.
4. Try and score one artery at a time. This will add time to the scoring portion of the test but it is easier for the eyes to become acclimated to the Coronary Anatomy when done individually. It has been found that score repeatability increases when using this technique. Good scoring repeatability is essential for accurate patient results.
5. If scoring by clicking on the region of calcium, make sure to click in the middle of the calcium so the system fills in the calcium completely. Also verify that the calcium has been completed filled in by looking at the region of interest. If the calcium does not fill in completely, make a box around the area of calcium and it should fill in.

PATIENT REPORT

1. The defaults for the patient report are as follows: Patient Demographics; Name, Patient ID, Exam Date, Sex, and Age are on, Blood Pressure, HDL, LDL and Triglycerides are off.
General : Image Output – Selected images, Images per sheet – 6, Heart Diagram – off. The percentile ranking is selected on.
2. If the percentile ranking is defaulted on and no information is entered into the ranking guide, a percentile ranking message will appear putting the patient into the 90th percentile even though the patient is not in the 90 percentile for their age. To avoid this make sure to fill in the percentile ranking table or deselect the table. Make sure to click on the Save to Default button, so it is saved to the patient report permanently on your system.
3. When the percentile ranking table is turned on in the preference screen, and the user scores a case, the system will look at the score and the percentile chart and pick the percentile which is valid for the patients age. For example if the patient is 52 years old and had a score of 91, and the default table had a value of 90 in the 50th percentile and 150 in the 75th percentile, the patients percentile would go to the 75th percentile.
4. After the initial information is typed into the patient report remember to click on “Save to default”. This will permanently save the information for all patient reports.
5. The patient report fields can be edited for the current patients without changing the defaults by selecting the “Apply button”. This will save the new information on the patient report for the current patient only. Once the user exits out of the current patient report, the default patient report information will be available.
6. In order to have the patient report print out with a score for each artery, score an artery even if there is no calcium in the artery. This will fill in the individual artery score table with a zero for each artery.
7. If you would like to have the arteries labeled when you print out the images on the patient report, score each artery by drawing a box around the artery and the label will appear above the artery.

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